



FIRST SEMESTER 2026-2027
Course Handout (Part II)

Date: 03/08/2026

In addition to part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No. : MATH F313
Course Title : Numerical Analysis
Instructor-in-charge: Devendra Kumar
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Chamber No. 6067-M (NAB))

1. Course Description:

Solution of non-linear algebraic equations; interpolation and approximation; numerical differentiation and quadrature; solution of ordinary differential equations; systems of linear equations; matrix inversion; eigenvalue and eigenvector problems; round off and conditioning.

2. Scope and objectives of the course:

Enables one to devise algorithms for the numerical solutions of mathematical problems. Also discusses the error analysis of different algorithms.

3. Course learning outcomes:

Upon successful completion of this course, students should be able to:

- **CLO1.** Explain the fundamental concepts of numerical analysis, including sources of computational errors, floating-point arithmetic, error propagation, and conditioning.
- **CLO2.** Apply appropriate numerical methods to determine the roots of nonlinear equations and evaluate the convergence, accuracy, and efficiency of different iterative algorithms.
- **CLO3.** Solve systems of linear and nonlinear equations using direct and iterative numerical techniques and analyze the stability and conditioning of computational methods.
- **CLO4.** Construct interpolation polynomials using standard interpolation techniques and estimate interpolation errors for function approximation.
- **CLO5.** Compute numerical derivatives and definite integrals using finite difference and quadrature methods and assess the associated approximation errors.
- **CLO6.** Obtain numerical solutions of ordinary differential equations arising from initial value problems using single-step and multistep numerical methods.
- **CLO7.** Solve boundary value problems using finite difference methods and interpret the numerical solutions obtained.
- **CLO8.** Determine dominant eigenvalues and eigenvectors of matrices using numerical algorithms such as the power and inverse power methods.
- **CLO9.** Apply weighted residual techniques, including the Rayleigh–Ritz, Collocation, and Galerkin methods, to obtain approximate solutions of boundary value problems and compare their applicability and effectiveness.





4. Method of conducting the course:

The instructional approach for this course combines theoretical concepts, analytical problem-solving,

and computational practice. Classroom lectures will introduce the fundamental principles, mathematical formulations, and derivations of numerical methods. Tutorial sessions will reinforce these concepts through guided problem-solving, analytical exercises, and discussion of practical applications. Selected numerical methods will be implemented using MATLAB to provide hands-on computational experience, facilitate visualization, and enhance problem-solving skills. Student learning will be assessed continuously through quizzes, assignments, laboratory/computational exercises, and examinations to ensure the development of both conceptual understanding and practical proficiency.

5. Textbook: Applied Numerical Analysis by Curtis F. Gerald & Patrick O. Wheatley, Pearson Education (7th ed.) 2003.

6. Reference Books:

RB1. Numerical Analysis, Burden and Faires, 7th ed., Thomson Learning 2001.

RB2. Numerical Methods for Engineers and Scientists, J. D. Hoffman, 2nd ed., CRC 2010.

RB3. A Friendly Introduction to Numerical Analysis, Brian Bradie, Pearson Education 2006.

RB4. Elementary Numerical Analysis, SD Conte & Carl de Boor, 3rd ed., TMH 2006.

RB5. Introduction to Numerical Analysis, Devi Prasad, 3rd ed., Narosa 2006.

7. Course Plan: Section numbers below are from textbook TB, unless otherwise specified.

Module	Topics	Section No.	Learning Outcome
1.	L1: Introduction of numerical analysis, using a computer to do numerical analysis, Errors, Sources of errors, and Floating-point arithmetic L2: Arithmetic accuracy in computers, Significant digits, Relative error, Propagation of errors, Avoiding loss of significant digits L3: Evaluation of polynomial T1: Related exercise problems of L1-L3	0.1-0.6	To understand the potential pitfalls of numerical computing
2.	L4: Non-linear equations, Bisection, and Secant methods to find the roots of non-linear equations L5: Method of false position L6: Newton's method T2: Related exercise problems of L4-L6 L7: Convergence of Newton's method L8: Fixed point iteration method, Order of convergence L9: Multiple roots T3: Related exercise problems of L7-L9	1.1-1.3, 1.5, 1.6	To find the roots of non-linear equations and understand the relative strengths and weaknesses of each method for system of non-linear equations





3.	L10: System of linear equations, Gaussian elimination method, LU decomposition L11: Pivoting, Scaling, Other direct methods L12: Pathology in linear systems - Singular matrices, Determinants and Matrix inversions T4: Related exercise problems of L10-L12 L13: Tri-diagonal systems, Thomas algorithm, Norms, Condition numbers, and Errors in computed solution L14: Jacobi, Gauss-Seidel methods L15: Newton and Fixed-point methods for nonlinear systems T5: Related exercise problems of L13-L15	2.1-2.5, 1.7	To solve a linear system using Gaussian elimination and iterative methods To compute the inverse of a matrix and understand the relative strengths and weaknesses of each computational method
4.	L16: Existence and Uniqueness of interpolating Polynomials L17: Lagrange polynomials and the Divided Differences method L18: evenly spaced points, Newton's forward and backward formulas T6: Related exercise problems of L16-L18 L19: Osculating & Hermite interpolating polynomials L20: Error of interpolation, Interpolation with repeated abscissa T7: Related exercise problems of L19-L20	3.1, 3.2, 4.4 (R5)	What is an interpolating polynomial and how to evaluate it efficiently?
5.	L21: Derivatives from difference table L22: Higher order derivatives L23: Newton-Cotes integration formulas, The Trapezoidal rule-a composite formula T8: Related exercise problems of L21-L23 L24: Simpsons rule L25: Error in numerical integration 26: Gaussian quadrature L27: Gauss Chebyshev and other methods T9: Related exercise problems of L24-L27	5.1-5.3, 5.6	To compute numerical derivatives and integration using discrete data points and to know how to integrate functions containing singularities
6.	L28: Taylor series method L29: Euler and Modified Euler's methods L30: Runge-Kutta (RK) methods T10: Related exercise problems of L28-L30 L31: Multistep methods: Milne's method L32: Adams-Moulton method, Predictor-corrector formulas L33: System of equations and Higher-order equations T11: Related exercise	6.1-6.5	To compute numerical solutions of initial value problems





	problems of L31-L33		
7.	L34-35: Boundary value problems (BVPs), Finite difference method L36: Solution through a set of equations, Derivative boundary conditions T12: Related exercise problems of L34-L36	6.7	To solve boundary value problems
8.	L37: Power method, Inverse power method of finding eigenvalues and eigenvectors of matrices	6.8	How to find dominant eigenvalues and eigenvectors
9.	L38: The Rayleigh-Ritz method L39: The Collocation method L40: The Galerkin method T13: Related exercise problems of L37-L40	9.1-9.2 (R2)	To apply the Rayleigh-Ritz, Collocation, and Galerkin methods to obtain approximate solutions of boundary value problems and compare their principles and applications.

8. Evaluation Scheme:

S. No.	Evaluation components	Weightage (Marks)	Duration (Minutes)	Date & Time	Nature of component (Closed Book/ Open Book)
1.	Mid-semester exam	35% (70)	90	As announced by AUGSD	Open book
2.	Quizzes (1 Surprised and 3 Announced, best score of three quizzes will be considered out of 4 quizzes)	20% (40)	20 minutes each	Announced Quizzes Dates: Q1: 21/08/2026 Q2: 25/09/2026 Q3: 20/11/2026	Closed book
3.	Comprehensive exam	45% (90)	180	As announced by AUGSD	Closed book

9. Make-up Policy: Make-up for mid-semester and comprehensive exams will be given only for very genuine cases, and prior permission must be obtained from IC. **No makeup will be given for quizzes under any circumstance whatsoever.**

10. Attendance Policy: Each student is expected to attend all classes and appear in all evaluative components.

11. Cut off for NC grade: Any student scoring less than 10% of the total marks (i.e., less than 20





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marks out of 200) will be given NC grade. Further, if a student is absent in mid-semester or comprehensive exam or in all four quizzes, then he/she will be given NC grade.

12. Chamber consultation hour: Wednesday 10 AM – 11 AM

13. Notices: The notices concerning this course will be displayed on the Nalanda website.

Instructor-in-charge
MATH F313

